I Overview

Previously we looked at a time-varying optimal control feedback law. But when we did pole placement we got a constant feedback.

↳ If all we care about is stability as $t \to \infty$ then (Asymptotic property) we can optimize this by

$$\min_{u(\cdot)} \int_{t_0}^{\infty} (x^T(t) Q x(t) + u^T(t) R u(t)) \, dt$$

$$Q \geq 0 \quad R > 0$$

This is important because if (A,C) is not C.C. then our solution will not be finite

Required for stability

No terminal cost; $J < \infty$, we need $x(t) = 0$ as $t \to \infty$

Q must be chosen so that all states are represented in the cost

↳ $Q = C^T C$ } (A, C) must be observable } (A,B) C.C

↳ Nothing to do with system sensors or output. This is a new C. We design this C.

II Solution

We must have a steady-state solution

$$\dot{P} = 0$$

Steady state occurs as $t \to \infty$

A) Fact: if (A, B) is C.C. the infinite horizon problem has a unique solution

$$u = -R^{-1} B^T P x$$

where P satisfies Algebraic Riccati Equation

$$\dot{P} = 0 = A^T P + PA + Q - PBR^{-1}B^T P$$

where $u = -Kx$ where $K = R^{-1} B^T P$

B) "Optimal" Poles

$$\text{eig}(A - B\underbrace{R^{-1}B^T P}_{K}) \qquad \dot{x} = Ax + Bu = (A - BK)x$$

III Optimal Output Feedback

What if we care about certain states instead of full state

↳ Again output $y = Cx$ has nothing to do with this

$$\min_{u(\cdot)} = J(u(\cdot)) = \int_{t_0}^{\infty} z^T(t)\, Q\, z(t) + u^T(t) R u(t)\, dt$$

$$\Rightarrow \min_{u(\cdot)} J(u(\cdot)) = \int_0^{\infty} x^T(t) \underbrace{G^T Q G}_{\hat{Q}} x(t) + u^T(t) R u(t)\, dt$$

$$\Rightarrow \min_{u(\cdot)} J(u(\cdot)) = \int_0^{\infty} x^T(t)\, \hat{Q}\, x(t) + u^T(t) R u(t)\, dt$$

$$\underbrace{\hat{Q} \geq 0 \quad \hat{Q} = C_1^T C_1 \quad \& \quad (A, C_1) \text{ is C.O.}}_{\text{Necessary and sufficient}}$$

A) Sufficient condition

$Q > 0 \quad (A, G)$ is C.O.

B) The Input becomes

$$\left.\begin{aligned} u &= R^{-1} B^T P x \quad P \text{ satisfies ARE} \\ 0 &= A^T P + PA + \hat{Q} - PBR^{-1}B^T P \end{aligned}\right\} \text{This is still full state feedback control}$$

↳ Cost is defined in terms of the output

IV Example

$$\min_{u(\cdot)} \int_0^\infty (4x_1^2 + 5x_2^2 + u^2)\, dt$$

s.t. $\dot{x}_1 = x_2 \quad \dot{x}_2 = u$

$$x = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \qquad \dot{x} = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u$$

$$Q = \begin{bmatrix} 4 & 0 \\ 0 & 5 \end{bmatrix} \qquad R = 1$$

Is Q well formed?

$Q \succ 0$

$$Q = C_c^T C_c \Rightarrow C_c = \begin{bmatrix} 2 & 0 \\ 0 & \sqrt{5} \end{bmatrix}$$

Showed in Q ways that Q is well formed

$$\begin{bmatrix} C_c \\ C_c A \end{bmatrix} = \begin{bmatrix} 2 & 0 \\ 0 & \sqrt{5} \\ 0 & 2 \\ 0 & 0 \end{bmatrix} \qquad \text{rank}(\Omega) = 2 \Rightarrow C.C.$$

Is the system controllable

$$[B \;\; AB] \Rightarrow \text{rank}(M) = 2 \Rightarrow C.C.$$

ARE: $u = -Kx \qquad K = R^{-1} B^T P$

$$A^T P + PA + Q - PBR^{-1}B^T P = 0$$

$$\begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} P_{11} & P_{12} \\ P_{12} & P_{22} \end{bmatrix} + \begin{bmatrix} P_{11} & P_{12} \\ P_{12} & P_{22} \end{bmatrix} \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} + \begin{bmatrix} 4 & 0 \\ 0 & 5 \end{bmatrix} -$$

$$\begin{bmatrix} P_{11} & P_{12} \\ P_{12} & P_{22} \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} [1] [0 \;\; 1] \begin{bmatrix} P_{11} & P_{12} \\ P_{12} & P_{22} \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

$$\Rightarrow P \in \left\{ \underset{1}{\begin{bmatrix} 6 & 2 \\ 2 & 3 \end{bmatrix}}, \underset{2}{\begin{bmatrix} -6 & 2 \\ 2 & 3 \end{bmatrix}}, \underset{3}{\begin{bmatrix} -2 & -2 \\ -2 & 1 \end{bmatrix}}, \underset{4}{\begin{bmatrix} 2 & -2 \\ -2 & -1 \end{bmatrix}} \right\}$$

$eig(1) = 7, 2$ $\quad eig(2) = -7, -2$ $\quad eig(3) = -3, 2$

$eig(4) = 3, -2$

The optimal control is then: $u = -R^{-1}B^T P x$

We solved for P above.

Optimal Poles are

$$eig(A - BR^{-1}B^T P) = \{-1, -2\}$$

↳ Close loop system is GAS

Solution using MATLAB

are $(A, B, R^{-1}B^T, Q) = \underline{u}$

or $lqr(A, B, Q, R) = K$

$\underline{u = -Kx}$

V Set-Point Control

We don't always go to zero

$\dot{x} = Ax + Bu \quad z = Gx \quad x \in \mathbb{R}^n \quad u \in \mathbb{R}^k \quad z \in \mathbb{R}^l$

Define a static reference that we would like our system to reach. ↳ $r \in \mathbb{R}^l$

$\bar{z} = z - r \qquad u = \underset{\text{keep } z \text{ at } r}{u_{eq}} + \underset{\text{converge to } r}{\bar{u}}$

Evaluate if an equilibrium point exists that will keep us at the desired point

$$Ax^{eq} + Bu^{eq} = 0$$

$$r = Gx^{eq}$$

$$z = Gx \text{ and } z^{eq} = r - Gx^{eq}$$

$$\begin{bmatrix} A & B \\ G & 0 \end{bmatrix} \begin{bmatrix} x^{eq} \\ u^{eq} \end{bmatrix} = \begin{bmatrix} 0 \\ r \end{bmatrix}$$

Assuming an equilibrium point exists, what are the dynamics of the new state.

$$\chi = x - x^{eq} \qquad \dot{\chi} = \dot{x} - \dot{x}^{eq} = Ax + Bu + \cancel{\dot{x}^{eq}}^{\,0}$$

$$\Rightarrow x = \chi + x^{eq} \quad \Rightarrow \dot{x} = A(\chi + x^{eq}) + B(u^{eq} + \bar{u})$$

$$\Rightarrow \underline{\dot{x} = A\chi + B\bar{u}} + \underbrace{\cancel{Ax^{eq} + Bu^{eq}}}_{\text{is 0 by definition}}$$

$$\bar{z} = Gx - r = G(\chi + x^{eq}) - Gx^{eq} = \underline{G\chi}$$

Formulate the optimal control Problem

$$\int_0^\infty (z^T(t) Q \bar{z}(t) + \bar{u}^T(t) R \bar{u}(t)\, dt$$

$$= \int_0^\infty \chi^T(t) G^T Q G \chi + \bar{u}^T R \bar{u}\, dt$$

Trivial if $Q \succ 0$ and (A, G) C.O.

We have: $R \succ 0$ $\quad Q \succeq 0$ $\quad (A, C)$ is C.O.

(A, B) is C.O., an equilibrium point exists.

(A, B) Given by dynamics

$$\hat{Q} = G^T Q G$$

$$u = -R^{-1} B^T P x = -Kx$$

$$u = u^{eq} - K(x - x^{eq})$$

P satisfies ARE

$$0 = A^T P + PA + \tilde{Q} - PBR^{-1}B^T P$$

Using MATLAB

$$K = lqr(A, B, \tilde{Q}, R)$$

VI Example

$$\dot{x} = \begin{bmatrix} -1/2 & -1/2 \\ 1/2 & 0 \end{bmatrix} + \begin{bmatrix} 1/2 \\ 0 \end{bmatrix} u$$

$$x_d = \begin{bmatrix} 0 \\ 2 \end{bmatrix} \quad r = 2 \qquad z = x_2 = [0 \quad 1]\, x$$

$$\underbrace{\begin{bmatrix} A & B \\ G & 0 \end{bmatrix}}_{\bar{A}} \begin{bmatrix} x^{eq} \\ u^{eq} \end{bmatrix} = \begin{bmatrix} 0 \\ r \end{bmatrix}$$

$$\begin{bmatrix} x^{eq} \\ u^{eq} \end{bmatrix} = \bar{A}^{-1} \begin{bmatrix} 0 \\ 0 \\ 2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 2 \end{bmatrix} \qquad x^{eq} = \begin{bmatrix} 0 \\ 2 \end{bmatrix}$$

$$u^{eq} = 2$$

Check observability (A, G)

$$\Omega = \begin{bmatrix} G \\ GA \end{bmatrix} \qquad \text{rank}(\Omega) = 2 \Rightarrow \text{C.O.}$$

Check controllability

$$\Gamma = [B \;\; AB] \qquad \text{rank}(\Gamma) = 2 \Rightarrow \text{C.C.}$$

We can now formulate

$$\int_0^\infty \bar{z}^T Q \bar{z} + \bar{u}^T R \bar{u} \, dt$$

$$= \int_0^\infty x^T G^T Q G x + \bar{u}^T R \bar{u} \, dt$$

To meet observability we need $Q > 0$

Design Choice:

$$Q = 10 \qquad r = 1$$

We could then find $\hat{Q} = G^T Q G = \begin{bmatrix} 0 \\ 1 \end{bmatrix} 10 \begin{bmatrix} 0 & 1 \end{bmatrix}$

$$\Rightarrow \hat{Q} = 10 \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} = C_c^T C_c$$

$$\Rightarrow C_c^T = \begin{bmatrix} 0 & 0 \\ 0 & \sqrt{10} \end{bmatrix}$$

State Feedback control

$$\bar{u} = -Kx \qquad K = \text{lqr}(A, B, \hat{Q}, r)$$

$$u = u^{eq} + \bar{u} = 2 - K\left(x - \begin{bmatrix} 0 \\ 2 \end{bmatrix}\right)$$